



BLY 111

GENERAL BIOLOGY 1

TOPIC: CELL STRUCTURE AND ORGANISM

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CELL

A cell is a membrane-bounded unit that contains the DNA hereditary machinery and cytoplasm. All organisms are cells or aggregates of cells.

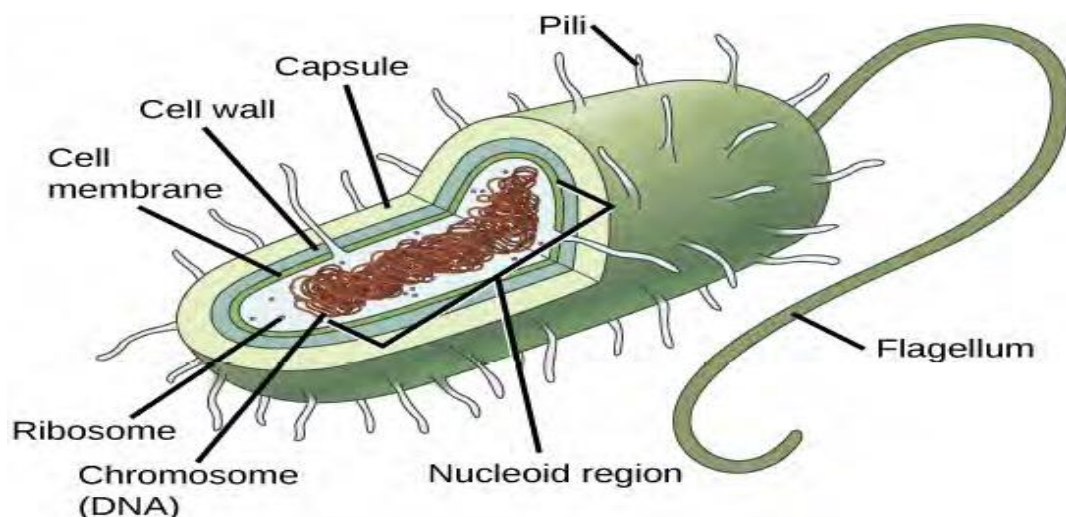
THE CELL THEORY

1. All organisms are composed of one or more cells, and the life processes of metabolism and heredity occur within these cells.
2. Cells are the smallest living things, the basic units of organization of all organisms.
3. Cells arise only by division of a previously existing cell. Although life likely evolved spontaneously in the environment of the early earth, biologists have concluded that no additional cells are originating spontaneously at present. Rather, life on earth represents a continuous line of descent from those early cells.

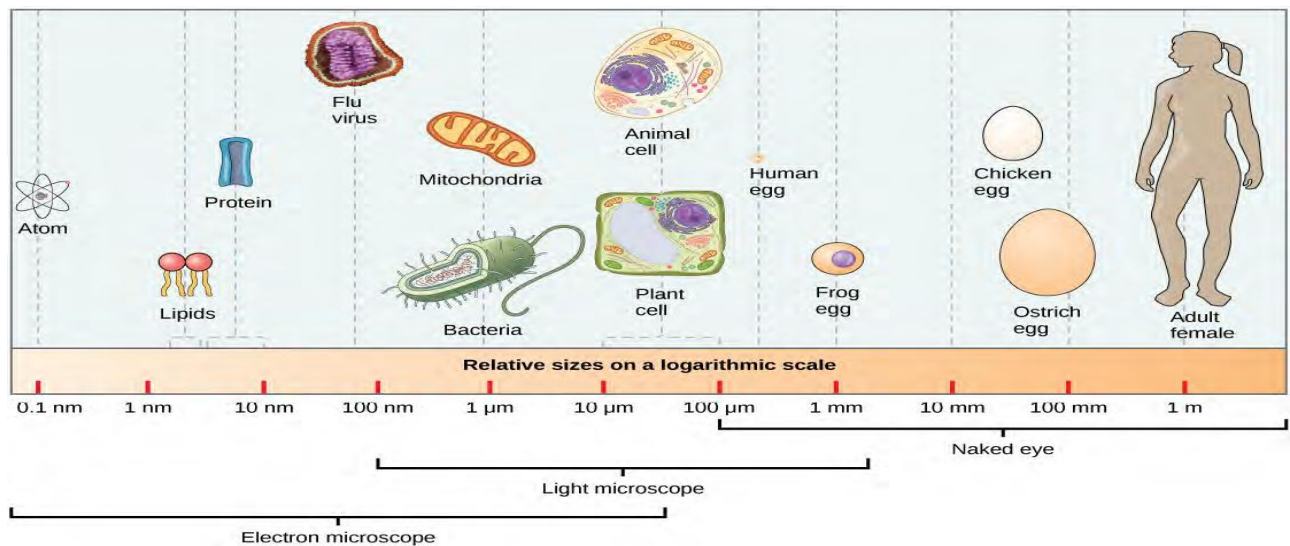
ALL CELLS COMMONALITY

- All cells share four common components
 - 1) a plasma membrane, an outer covering that separates the cell's interior from its surrounding environment
 - 2) Cytoplasm, consisting of a jelly-like cytosol within the cell in which other cellular components are found
 - 3) DNA, the genetic material of the cell
 - 4) Ribosomes, which synthesize proteins

COMPONENTS OF PROKARYOTIC CELLS



SIZES



SOME VITAL COMPONENTS OF PROKARYOTIC CELLS

1. CAPSULE;

A thick polysaccharide layer outside of the cell wall. Used for sticking cells together, as a food reserve, as protection against desiccation and chemicals, and as protection against phagocytosis. In some species the capsules of many cells fuse together forming a mass of sticky cells called a biofilm. Dental plaque is an example of a biofilm.

2. CELL WALL;

Made of murein (not cellulose), which is a glycoprotein (i.e a protein/carbohydrate Complex, also called peptidoglycan).

3. PLASMA MEMBRANE;

sometimes also called the "cell membrane" it is phospholipid bilayer. It is flexible i.e. non-rigid.

4. CYTOPLASM;

Prokaryote cytoplasm is a gel-like substance that consists mainly of water. In general cytoplasm may also contain enzymes, salts, cell components and various organic molecules.

5. FLAGELLUM;

A rigid rotating helical-shaped tail used for propulsion. The motor is embedded in the cell Membrane and is driven by an H⁺ gradient across

the membrane. The bacterial flagellum is quite different from the eukaryotic flagellum. The flagellum of a prokaryotic cell does not "beat" but rotates about a "bearing" in the cell wall - resulting in a "corkscrew" motion that drives the cell forwards as a propeller propels some ships forwards

6. **PILI –**

A hair-like appendage many of which are present on the surface of many bacteria. It is a protein rod that is sometimes called a proteinaceous appendage. Pili help bacterial cells adhere to each other, to animal cells and to inanimate objects. A single bacterium can have as many as 1,000 fimbriae.

7. **PHOTOSYNTHETIC MEMBRANES;**

Photosynthetic membranes are surfaces on which light-absorbing pigments are located. Bacteriochlorophylls are photosynthetic pigments found in phototrophic bacteria.

8. **MESOSOMES;**

Are structures of prokaryotic cells formed by folded invaginations (sometimes described simply as "infoldings") of the plasma membrane. Enzymes associated with respiration of the prokaryotic cell are located on the infoldings. Not all prokaryotic cells have mesosomes. Nuclear body (genetic material). The genetic material consists of a circular double-stranded DNA which is not associated with proteins to form chromatin.

9. **PLASMIDS;**

Plasmids are short pieces of circular DNA. Separate from the main DNA loop. They replicate independently of the cell genome, and used to exchange DNA between bacterial cells, and also very useful for genetic engineering. Plasmids are not present in eukaryotic cells.

10. **FOOD STORES;**

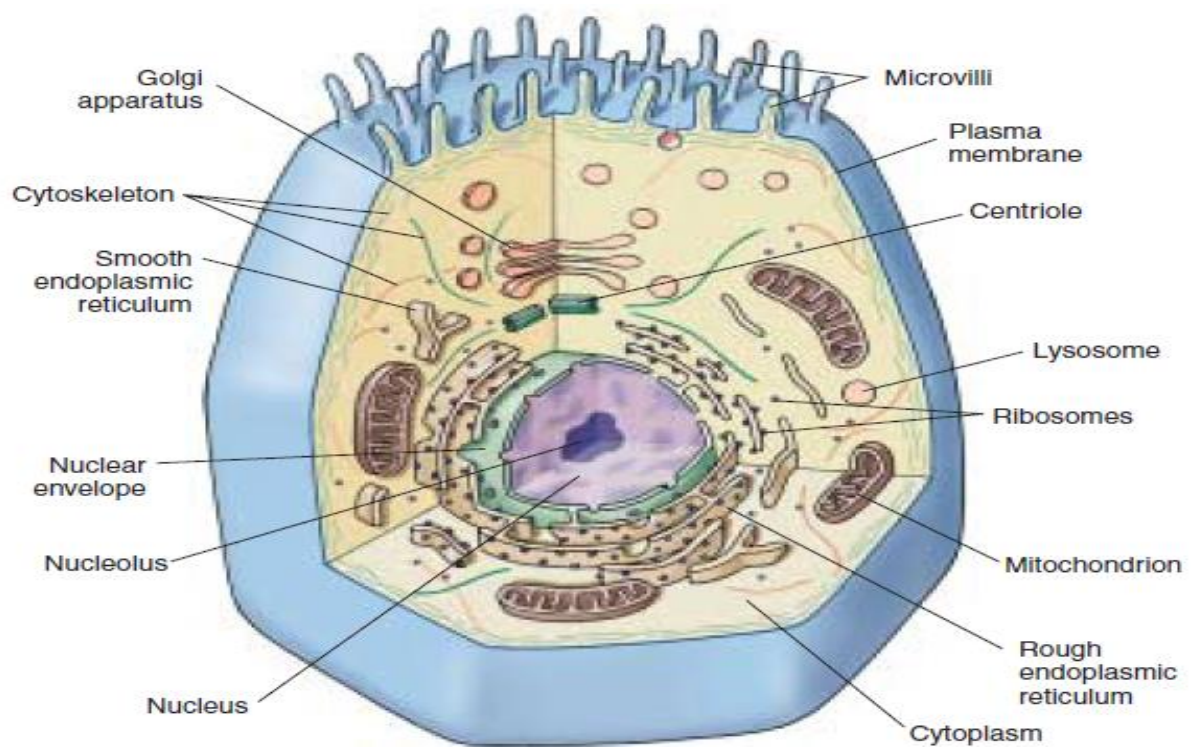
The food stores in prokaryotic cells can be lipid globules or glycogen granules.

11. RIBOSOMES;

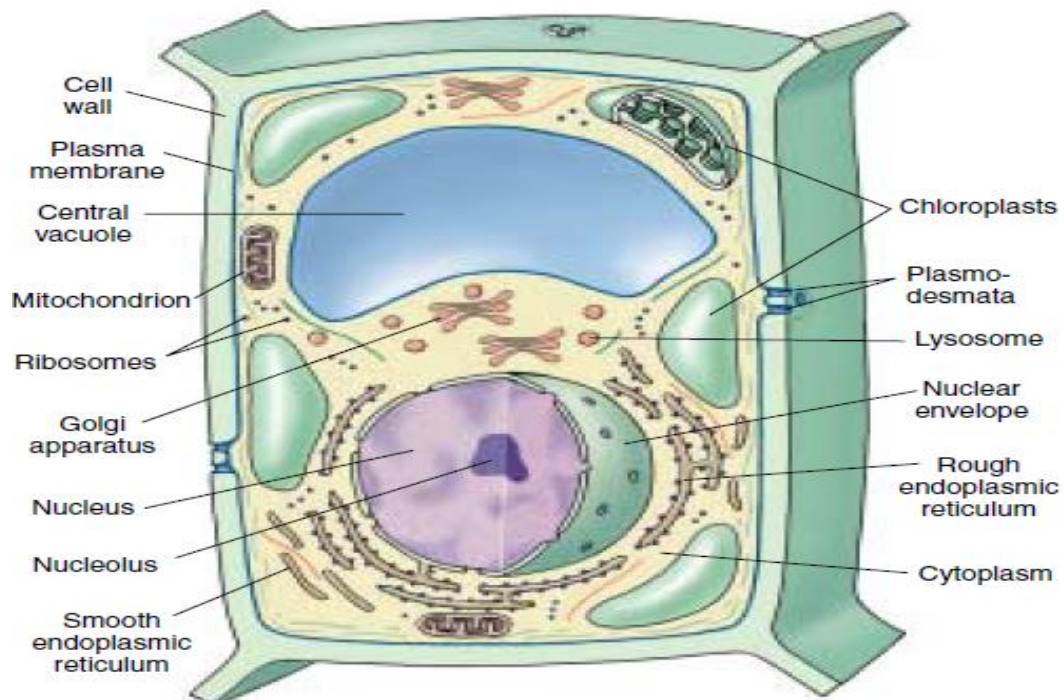
The ribosomes found in prokaryotic cells are much smaller than the ones in eukaryotic cells. The (smaller) ribosomes in prokaryotic cells are the 70S type (50+30).

EUKARYOTIC CELLS

- Protista, fungi, plants and animals are all composed of eukaryotic cells
- The hallmark of the eukaryotic cell is compartmentalization
- Eukaryotic cells contain membrane-bounded organelles that carry out specialized functions



A generalized diagram of an animal cell



FUNCTIONS OF EUKARYOTIC ORGANELLES

cell wall

- Plant cells have cell walls.
- The basic functions of plant cell walls are mechanical.
- Plant cell walls form part of a transport system known as apoplast system through which water and some solutes can pass through plant tissue using apoplastic pathways (along / through cell walls) and symplastic pathways (i.e. via the cytoplasm of a series of adjacent cells).

Plasma membrane

(also called the plasmalemma or the cell surface membrane)

Functions:

- Act as differentially-permeable surface, it controls movement of solutes in and out of the cell.
- It synthesizes and assemble cell wall components in plants
- The cell membrane maintains the physical integrity of the cell. Especially in animal cells (because they do not have cell walls).
- The cell membrane forms a barrier between the inside of the cell and the environment outside the cell

- The cell membrane physically separates the intracellular components (e.g. organelles in eukaryotic cells) from the extracellular environment.
- Cell membranes can allow active transport of specific molecules across the cell membrane in either direction, i.e. either into or out of the cell.
- The cell membrane protects the cell from some harmful chemicals in its external environment
- In many cases the cell membrane also helps to hold the cytoskeleton
- It helps in bulk transport (exocytosis and endocytosis).
- It serves as markers and helps in signaling.
- It partakes in Metabolic Activities.

Plasmodesmata

- They are tiny strands of cytoplasm that pass through pores in plant cell walls, forming "connections" or "pathways" between adjacent cells. They specifically form the symplast pathway for the movement of water and solutes through plant structures. These cell-cell connections are especially important for the survival of plant cells during conditions of drought

Cytoplasm

- It is the part of a plant cell that includes all the contents within the cell membrane but outside of the nucleus of the cell. It therefore includes the cytosol (i.e the semi-fluid part of a cell's cytoplasm) as well as the plant cell organelles. Also located within the cytoplasm is the cytoskeleton, which is a network of fibres whose function is to provide mechanical support to the cell, including helping to maintain the cell's shape. The cytoplasm therefore consists mainly of water and contains enzymes, salts, organelles, and various organic molecules. The cytoplasm helps to move materials around the cell and also dissolves cellular waste.

Vacuole

Each vacuole is enclosed by a vacuolar membrane called the **Tonoplast**

Content Of Vacuole

- Cell sap, which is a solution of salts, sugars and organic acids.
- Enzymes needed for recycling components of cells, e.g., chloroplast.
- Anthocyanins are sometimes present in cell vacuoles. These are chemical pigments responsible for some of the (non-green) colors of flowers, e.g., reds, blues, purples.

Functions:

- It maintains turgor pressure (turgidity) inside the cell. This pressure pushes the plasma membrane against the cell wall. Plants need turgidity to maintain rigidity.

Nucleus

- Nucleus is the "*control center*" of a eukaryotic cell (i.e., plant cells and animal cells). The nucleus controls the activity of the cell by regulating protein synthesis within the cell. The nucleus also contains genetic material and a nucleolus.

Nucleolus

- The nucleolus is located within the nucleus and is the site of synthesis of: Transfer RNA, Ribosomal RNA and ribosomal subunits

Nuclear Membrane

- The nuclear membrane is also called the nuclear envelope and encloses the contents of the nucleus of the cell - separating the contents of the nucleus from the rest of the cell. Nuclear pores in the nuclear membrane enable various substances, such as nutrients and waste products, to pass into and out of the nucleus.

Rough Endoplasmic Reticulum

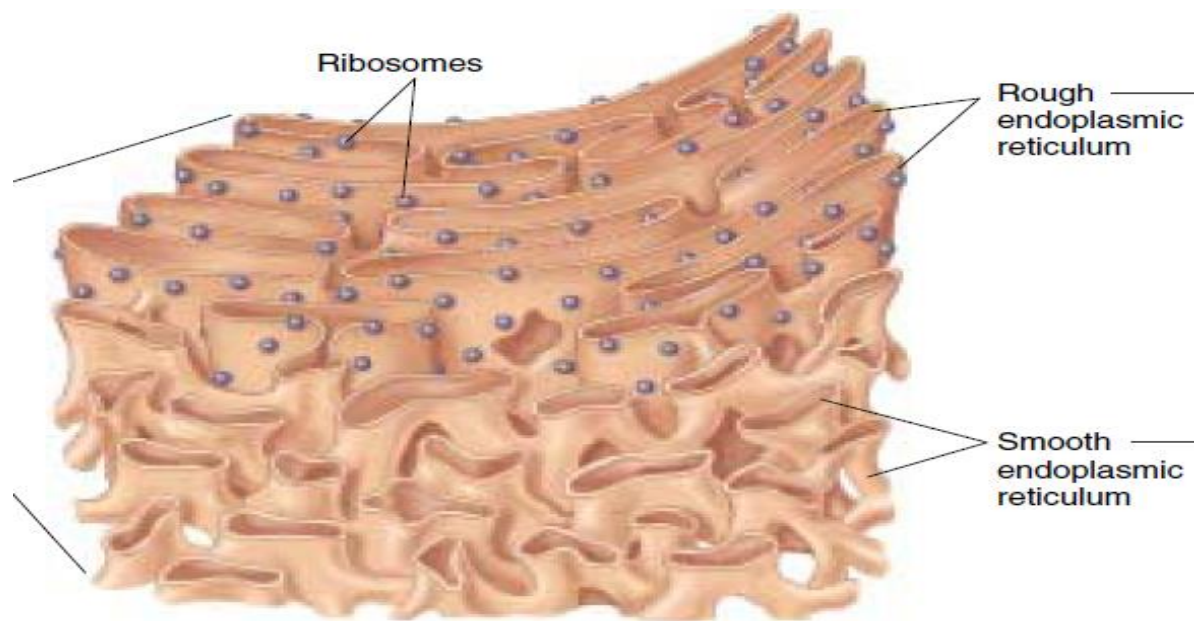
- Rough endoplasmic reticulum contain some ribosome specific transmembrane glycoprotein called ribophorins I and II help in attachment of ribosomes. It is the site of protein synthesis (which takes place within the ribosomes attached to the surface of the RER) as well as storage of proteins and preparation for secretion of those proteins. RER also synthesize membrane proteins and glycoproteins which are inserted into the rough endoplasmic reticulum membranes,

thus endoplasmic reticulum is the site of biogenesis of cellular membranes.

Smooth Endoplasmic Reticulum

- Is the site of lipid synthesis and secretion within cells, glycogenolysis and drug detoxification.

Endoplasmic Reticulum



Mitochondria

- Are structures found in both plant and animal cells. They are bounded by double membranes, the inner of which is folded inwards, forming projections (called cristae). The mitochondria matrix contains lipids, proteins, circular DNA molecule (mtDNA) and 55S ribosomes (mitoribosomes). They are considered semi autonomous organelle because they can synthesize 10% of their proteins.

Function of mitochondria is energy production.

- Mitochondria contain enzyme systems needed to synthesize adenosine triphosphate (ATP) by oxidative phosphorylation. Also,
- the inner membrane contain F_1 particles meant for ATP synthesis The quantity of mitochondria within cells varies with the type of cell.
- In the case of plant cells, mitochondria may be particularly abundant in sieve tube cells, root epidermal cells and dividing meristematic cells.

Chloroplasts

- They are the sites of photosynthesis within plant cells. They are very important parts of plant cells. The number of chloroplasts per cell varies according to the type of cell and its function.
- They are plentiful in leaf cells that receive sunlight - as opposed to root cells that do not receive light. They comprise of two envelopes, outer and inner.
- There occurs a third system of membranes within inner membrane called **grana** which is the **main functional unit of the chloroplast**.
- The grana is bathed in **homogenous matrix** called the **stroma**.
- Stroma contains a variety of photosynthetic enzymes.

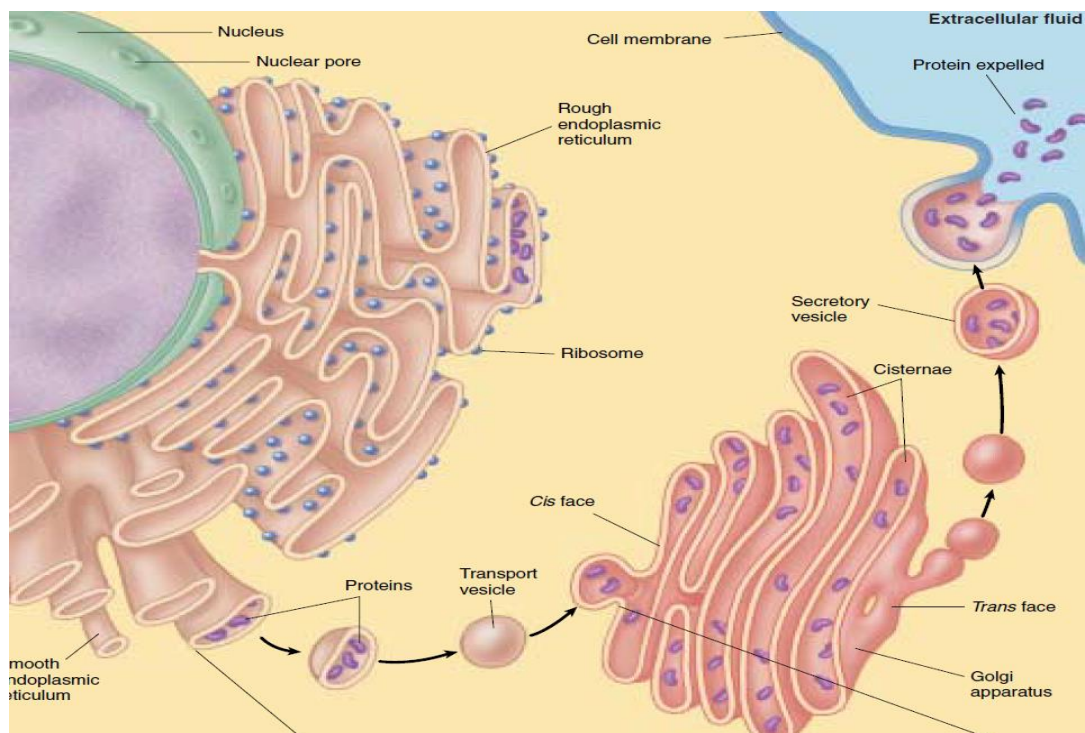
Chloroplasts are a type of plastid. There are also other types of plastids (not all of which are present in all plant cells but all of which are derived from proplastids). Others are chromoplast (phaeoplast and rhodoplast) Leucoplast (amyloplast, elaioplast and proteinoplast). All plastids contain small genome (DNA) and 70S ribosomes.

Golgi Body

- It consists of at least three distinct classes of cisternae: cis Golgi, median Golgi and trans Golgi, each has distinct enzymatic activities.

Golgi body performs the following functions:

- Packaging of secretory materials, such as enzymes, mucin, melanin pigment etc.
- The processing of proteins i.e phosphorylation, glycosylation, sulphation and selective proteolysis
- Synthesis of certain polysaccharide and glycolipids
- Sorting of proteins destined for various locations in the cell
- Proliferation of membranous element for the plasma membrane
- Formation of acrosome of sperms



Ribosomes

Ribosomes can either be "free" or "membrane-bound". They take part in the synthesis of some proteins by catalyzing the formation of those proteins from individual [amino acids](#) (using messenger RNA as a template). Examples of proteins catalyzed by ribosomes include glycoproteins, lysosome proteins, membrane proteins and some organelle proteins.

Ribosomes have sedimentation coefficient of about 80S comprising of two subunits called 40S and 60S. The smaller 40S subunit contain one molecule of 18S rRNA and 30 proteins (named as S₁, S₂, S₃ to S₃₀).

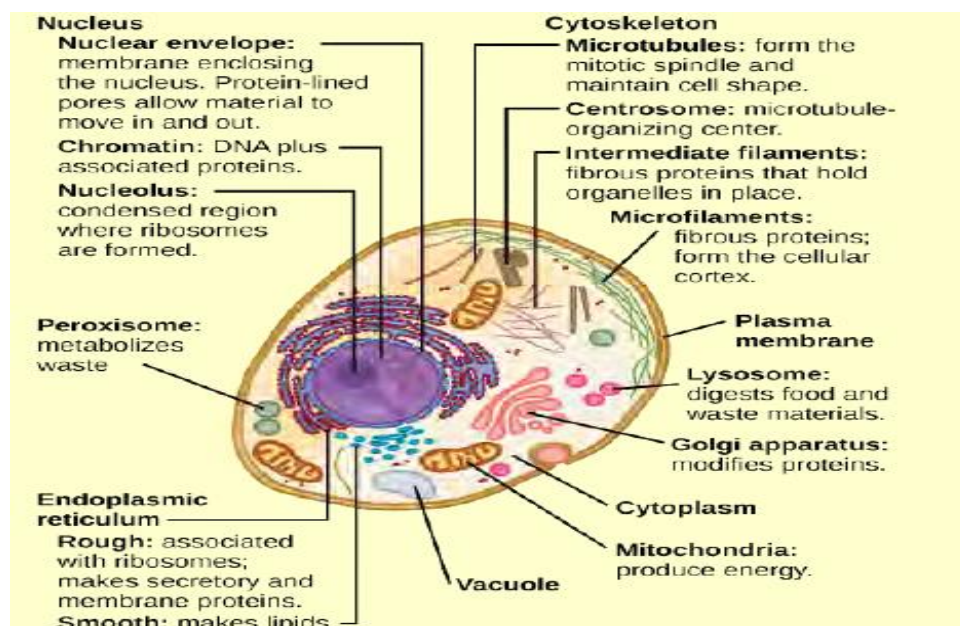
The larger 60S ribosomal subunit contains three types of rRNA molecules (28SrRNA, 5.8SrRNA, and 5SrRNA) and 40 proteins (named as L₁, L₂, L₃ to L₄₀).

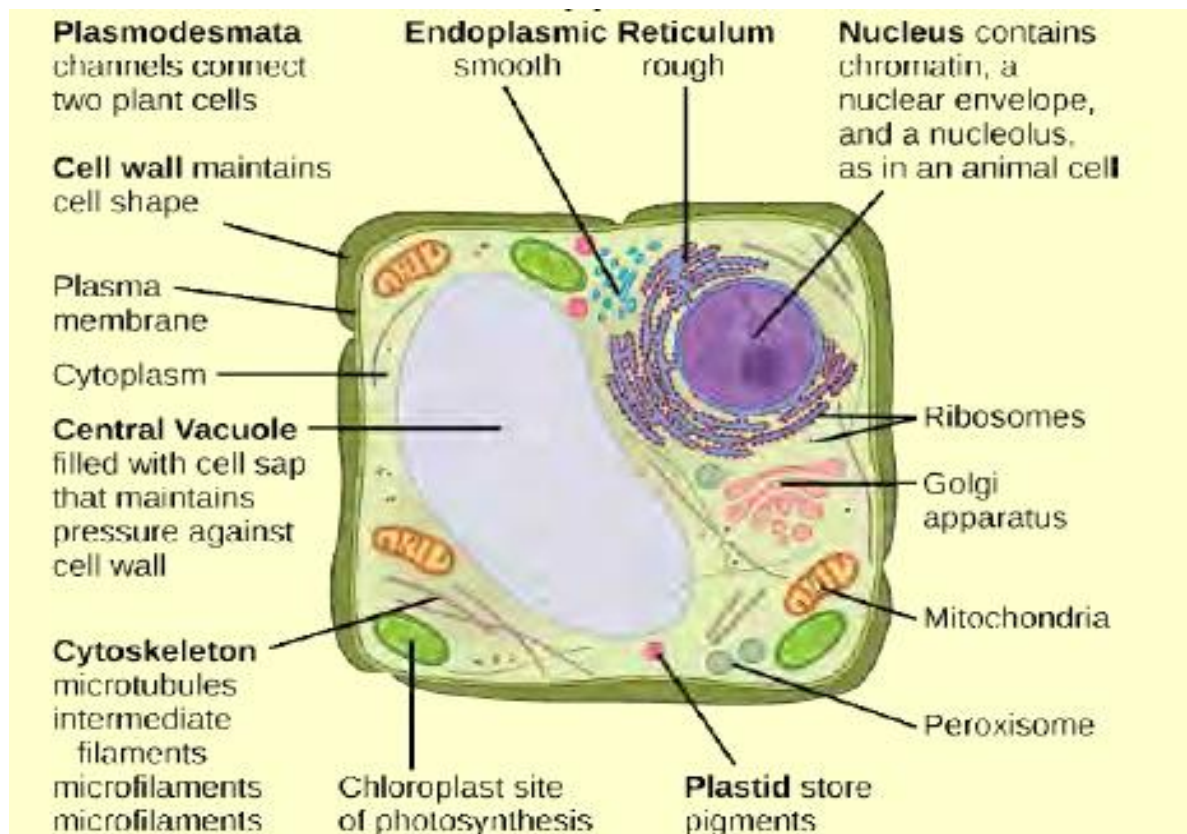
Lysosomes

They originate from golgi bodies and contain numerous hydrolytic enzymes. There are four types

- Primary lysosomes (storage granules)
- Secondary lysosomes (digestive vacuoles)
- Residual bodies
- Autophagic vacuoles

Lysosomes have high acidic medium of pH 5, they digest materials taken in by endocytosis. And can also digest the cell or worn-out cells (autophagy).

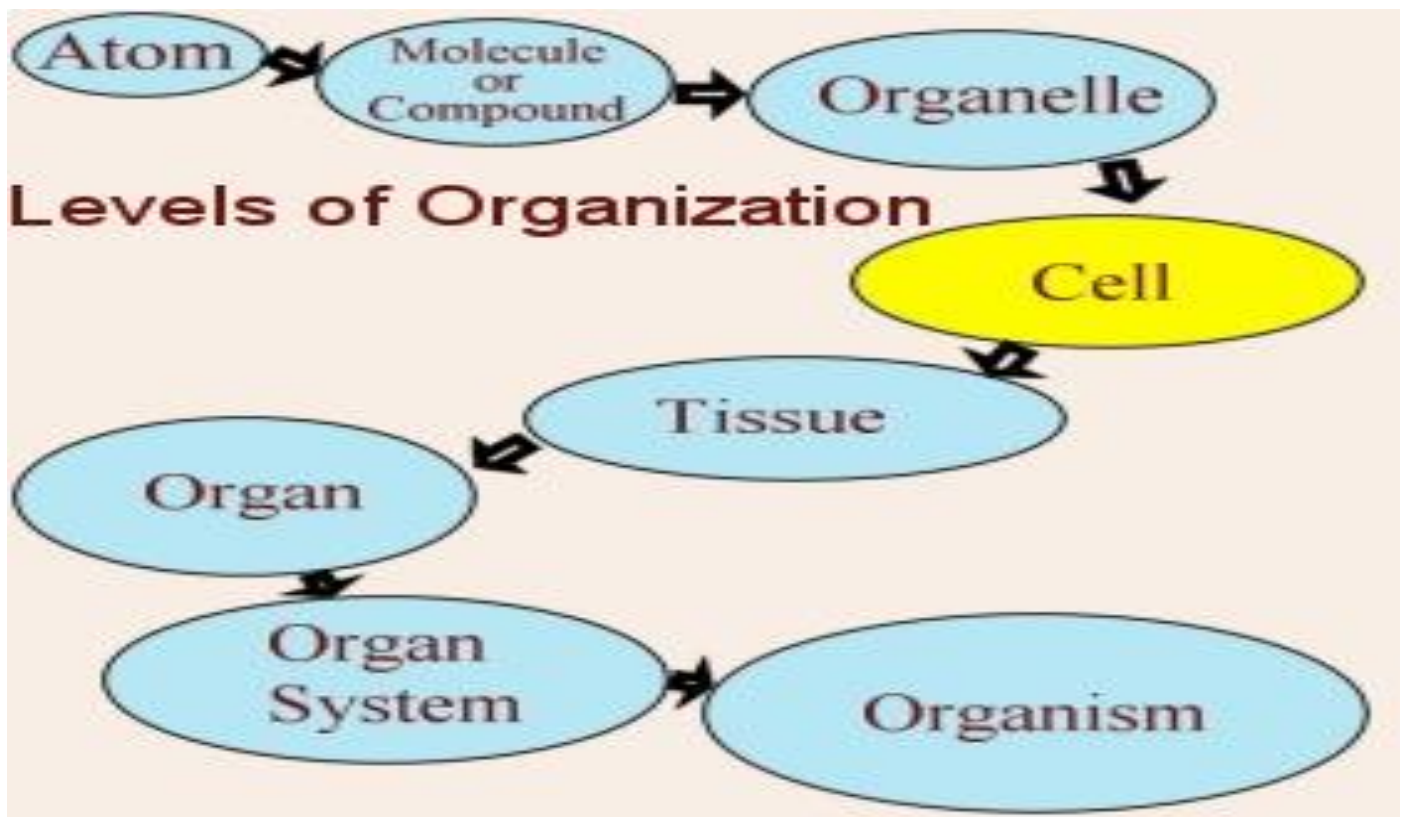




Eukaryotic Cell Structures and Their Functions

Structure	Description	Function
Cell wall	Outer layer of cellulose or chitin; or absent	Protection; support
Cytoskeleton	Network of protein filaments	Structural support; cell movement
Flagella (cilia)	Cellular extensions with 9 + 2 arrangement of pairs of microtubules	Motility or moving fluids over surfaces
Plasma membrane	Lipid bilayer with embedded proteins	Regulates what passes into and out of cell; cell-to-cell recognition
Endoplasmic reticulum	Network of internal membranes	Forms compartments and vesicles; participates in protein and lipid synthesis
Nucleus	Structure (usually spherical) surrounded by double membrane that contains chromosomes	Control center of cell; directs protein synthesis and cell reproduction
Golgi apparatus	Stacks of flattened vesicles	Packages proteins for export from cell; forms secretory vesicles
Lysosomes	Vesicles derived from Golgi apparatus that contain hydrolytic digestive enzymes	Digest worn-out organelles and cell debris; play role in cell death
Microbodies	Vesicles formed from incorporation of lipids and proteins containing oxidative and other enzymes	Isolate particular chemical activities from rest of cell
Mitochondria	Bacteria-like elements with double membrane	"Power plants" of the cell; sites of oxidative metabolism
Chloroplasts	Bacteria-like elements with membranes containing chlorophyll, a photosynthetic pigment	Sites of photosynthesis
Chromosomes	Long threads of DNA that form a complex with protein	Contain hereditary information
Nucleolus	Site of genes for rRNA synthesis	Assembles ribosomes
Ribosomes	Small, complex assemblies of protein and RNA, often bound to endoplasmic reticulum	Sites of protein synthesis

Levels of Organization



The levels of biological organization in order from smallest to largest are:

atom → molecule → macromolecule → organelle → cell → tissue → organ → organ system → whole organism.

TISSUES

Groups of similar cells that work together to perform a specific function.

4 major tissue types in animals

- Epithelial tissue
- Connective tissue
- Muscle tissue
- Nervous tissue

ORGANS

structures made of different types of tissues that work together to perform a specific function.

- **Examples**

Heart, Lungs, Stomach, Small intestine, Liver, Large Intestine, Gall Bladder, Plant Roots, Plant Stems, Plant Leaves

ORGAN SYSTEMS

Groups of organs that work together to perform a specific function.

- **Examples:**

Digestive system, Circulatory system, Respiratory system, Nervous system, Muscular system, Skeletal system, Integumentary system (skin), Vascular system in plants

ORGANISM

A complete, individual living thing.

- **Examples:**

A single person, A single plant, A single bacterium, A single protist

FORMS IN WHICH CELLS EXIST

- Independent / free living – only possess one cell
- Filament – joining of identical cells end to end. Spirogyra, Zyguema, Oscillateria and Oedogonium
- Colony – similar cells joined together, cannot be differentiated from each other. vovox
- Part of living organism- A group of cells form tissue, a group of tissues form organ, a group of organs form system, living organisms are made up of systems

The simplest living multicellular organisms

- [sponges](#), are made of many specialized types of cells that work together for a common goal. Such cell types include digestive cells, tubular pore cells, and epidermal cells. Though the different cell types create a large, organized, multicellular structure — the visible sponge — they are not organized into true interconnected tissues. If a sponge is broken up by passing it through a sieve, the sponge will reform on the other side. However, if the sponge's cells
- are separated from each other, the individual cell types cannot survive alone

Structure	Description
Organelle	Cell structure that is specialized to carry out a particular function or job
Cell	Basic structural and functional unit of a living organism
Tissue	Group of cells with similar structures, working together to perform a shared function
Organ	Structure made up of a group of tissues, working together to perform specific functions
Organ system	Group of organs with related functions, working together to perform body functions